Syllabus

MAT 5173 — Algebra I Fall 2026 · UT San Antonio

Texts and Materials

Primary text

- **Knapp** — Knapp, *Basic Algebra*. Digital Second Edition is free from the author and licensed for course posting.

Secondary reading

- **Rosebrock** — Rosebrock, *Visual Group Theory*
- **Beardon** — Beardon, *Algebra and Geometry*. The concrete companion: Ch. 1-3, 12, 13, 14.
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*. Chapter 1 only. No Groebner bases.

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*. Section 1.1 only — plane curves.
- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

Handouts & papers

- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*. Not held by UTSA. Chapter 1 (13 pp.) distributed as a handout.
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*. Notices of the AMS 55 (2008), no. 10. Co-reading for the Mobius lecture.

Course Information

Catalog Entry

MAT 5173 Algebra I

The opportunity for development of basic theory of algebraic structures. Areas of study may include monoids, semigroups, groups, isomorphism theorems, free groups, group extensions and group actions, Sylow theorems, group chains and composition series, nilpotent and solvable groups, cohomology of groups.

Prerequisites

MAT 4233 or consent of instructor.

Credit Hours

3

Course Modality

Traditional in-person

Class Meetings

Duration	Wed 19 Aug – Thu 3 Dec
Campus	Main Campus
Location	MH 3.03.10
Time	Mon Wed 7:30PM–8:45PM

Learning Objectives

By the end of this course you should be able to:

- Define the notions of group, cyclic group, subgroup, normal subgroup, homomorphism, isomorphism, permutation group.
- Describe and analyze examples of groups, emphasizing groups of geometric transformations.
- Define group actions, emphasizing geometric actions.
- Define the notions of ring, subring, domain, field, ideal, homomorphism, isomorphism.
- Define rings of polynomials over a field, the notion of degree, and irreducible factors.
- Explore connections of the abstract theory of rings, polynomials and ideals with the geometry of algebraic varieties.
- Communicate concepts in writing and debate technical arguments orally.
- Use a computer algebra system for numerical and visual explorations involving algebraic and geometric structures.

About the Instructor

Eduardo Dueñez, Associate Professor of Mathematics.

Department	Mathematics
Office	FLN 4.01.11
Student hours	Mon 5:30PM – 6:30PM & Tue 1:30PM – 2:30PM
Email	eduardo.duenez@utsa.edu
Homepage	https://supernumero.us/about

Email is the preferred method of communication.

Assessment and Grading

Weight of course activities

Component	Weight	
Q&A sessions	40%	5 sessions, 8% each
Problem sets	15%	5 sets, 3% each
Midterm Exam 1	20%	Wed 30 Sep in class
Midterm Exam 2	25%	Wed 2 Dec in class

Q&A sessions

Five class meetings throughout the semester (roughly every other Wednesday except when adjacent to an exam) will be *Q&A sessions* in which each student will be asked to explain and defend their solution to a homework problem submitted as part of a written assignment. You will only be asked questions related to an assigned problem and about the solution you submitted.

Each Q&A session is scored 0–3:

- **3** Correct explanation to a correct solution, demonstrating full understanding of the concepts and all aspects of the solution submitted.
- **2** A completely correct solution submitted is explained to an incomplete degree; **Or**: an unfinished solution, with a thorough and articulate account of what you tried, explaining where and why you got stuck while demonstrating knowledge of the background and reading necessary to solve the problem.
- **1** Correct solution inadequately explained revealing little or no understanding; *or*: partial solution accompanied by an incomplete but partly correct oral explanation.
- **0** Nothing submitted; *or*: work submitted (even if fully correct) is backed by no cogent explanation nor knowledge of concepts involved.

A correct answer you cannot explain is worth less than an explainable honest failure.

Note: A Q&A average ≤ 1.0 automatically caps the course grade to the level of 'D' (at most).

Problem sets

Five sets, each with three or four theoretical problems plus one SageMath computer exploration. **Maximum three pages.** Do *not* submit code; include the relevant snippets and output as part of your written submission. (SageMath problems are also included in Q&A.)

Each item will be scored *only* based on the **degree of completion** (not on correctness): **2** for a nominally complete solution, **1** for a clearly partial one, **0** for nothing submitted or for obvious filler with no substance.

Note: Any submission containing non-human-readable content, stray characters, or artifacts revealing non-proofread AI output scores 0 for that item.

Use of Artificial Intelligence (AI)

AI assistance is allowed in preparing work for submission, as are any solution keys, collaboration, and internet sources. You are ultimately responsible for your submissions; any part of your work that has clearly not been proofread (e.g., includes non-human readable characters or LLM artifacts) automatically earns a zero grade. *Include one line noting what resources you used in your submission.*

This policy is by design, and it reflects how the course is graded: 85% of the grade comes from Q&A sessions and in-class exams where only what is truly understood and can be explained counts. Assignments submitted but not written/understood by you result in failing grades in Q&A sessions and exams. Use AI to learn faster —not to prepare and turn in papers you cannot defend, blocking you from passing exams as well.

All audiovisual (or audio only, or video only) recordings of lectures are explicitly forbidden; all uses of AI agents or apps to capture, process, or analyze any lecture contents are **strictly forbidden**. The only *limited* exceptions are *explicitly* allowed recordings per official memorandum of the office of Student Disability Services. Under no circumstances is any processing or analyzing of such recordings allowed. Any sharing of confidential course materials with individuals who are not enrolled students in the course is a violation of FERPA privacy laws.

Exams

Both exams are traditional in-class (written and closed book). Exams are based **primarily on the assigned problem sets** —expect some context and phrasing to differ from the homework, plus a minority of other unseen problems. Memorizing solutions will not result in good exam grades; preparing by understanding the assigned problems will make the exams straightforward and earn high Q&A grades as well.

Attendance and missed work

No extensions on problem sets. A *single* missed Q&A session in the semester may be made up during office hours within seven days (no documentation is required). *No Q&A scores are dropped.*

Final grade ranges

Grade	Range	Grade	Range
A	[90%, 100%]	A-	[85%, 90%)
B+	[80%, 85%)	B	[76%, 80%)
B-	[72%, 76%)	C+	[69%, 72%)
C	[65%, 69%)	C-	[60%, 65%)
D	[50%, 60%)	F	[0%, 50%)

Time Commitment Expectations

Expect an average of about **9 hours per week**, and no fewer than 7.

- Attend class meetings (**3 hours per week**)
- Reading the assigned textbook sections (**2–3 hours per week**)
- Problem sets and the Sage exploration (**2–3 hours per week**)
- Office hours and study group (**1 hours per week**)

Additional Course Information

Each student should be intimately familiar with the contents of any work submitted for grading. The instructor has the right to request adequate **verbal explanation** of methodology and content for any submitted work. Credit will not be awarded when such an explanation is requested but not satisfactorily provided.

Course Schedule

Date	Topic
Wed 19 Aug	Lecture — Course launch; the plane and its isometries
Mon 24 Aug	Lecture — Reflections generate everything
Wed 26 Aug	Lecture — From symmetry to the group axioms
Mon 31 Aug	Lecture — Cyclic and dihedral groups
Wed 2 Sep	Lecture — Permutations and the sign homomorphism
Wed 9 Sep	**Q&A** (**Due**: Set 1)
Mon 14 Sep	Lecture — Subgroups; Cayley graphs
Wed 16 Sep	Lecture — Cosets and Lagrange

Date	Topic
Mon 21 Sep	Lecture — Homomorphisms and quotient groups
Wed 23 Sep	Lecture — Group actions: orbits and stabilizers
Mon 28 Sep	Lecture — Orbit-stabilizer and counting
Wed 30 Sep	**Q&A** (**Due**: Set 2)
Mon 5 Oct	**Exam** — Midterm Exam 1 — Sets 1 and 2
Wed 7 Oct	Lecture — Symmetries of the polyhedra; finite rotation groups in space
Wed 14 Oct	Lecture — Finite groups; frieze and wallpaper patterns
Mon 19 Oct	Lecture — Mobius transformations as a group action on the sphere
Wed 21 Oct	Lecture — Rings and polynomial rings k[x], k[x,y]
Mon 26 Oct	Lecture — Integral domains and fields of fractions; polynomials in one variable
Wed 28 Oct	Lecture — Ideals; prime and maximal ideals
Mon 2 Nov	**Q&A** (**Due**: Set 3)
Wed 4 Nov	Lecture — Affine space and affine varieties V(S)
Mon 9 Nov	Lecture — I(V) and the algebra-geometry dictionary
Wed 11 Nov	Lecture — The coordinate ring k[x,y]/I(V) — the landing
Mon 16 Nov	Lecture — Why C and not R: the Nullstellensatz, stated
Wed 18 Nov	**Q&A** (**Due**: Set 4)
Mon 23 Nov	Lecture — Plane curves — rational, cusp, node
Mon 30 Nov	*Review* — Review — the exam pool
Wed 2 Dec	**Exam** — Midterm Exam 2 — Sets 3 and 4

Readings for each meeting are on the Weekly Schedule (also available as a printable PDF).

Department and University Policies

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu.

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law.

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes or corrections to the course materials, assignment dates, or other updates will be communicated to students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Essential Student Information

- **Important:** Bookmark and visit the Common Syllabus Information webpage for resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit Academic Innovation's Student Technical Support page.
- UT San Antonio provides reasonable accommodations to students via Student Disability Services, or call (210) 458-4157.
- Your well-being is important. Support, including 24/7 mental health services, can be accessed from the Well-being at UT San Antonio site.
- Student Assistance Services offers confidential support and personalized guidance.
- The Roadrunner Pantry provides free access to groceries, toiletries, and other essentials.
- Students are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Review Section 203 of the Student Code of Conduct.
- Visit UT San Antonio Libraries and Museums for journals, research tutorials, and your department's librarian.